Momentum Conservation: Adding Up a Quantity That Refuses to Change

SYNOPSIS

Note 8 made a claim that sounds too strong to be true: add up the momentum of everything involved before an event, add it up again afterwards, and the two totals are equal — always, with no exceptions anyone has ever found. This lab is where you stop taking that on trust. Two carts on a level track interact three times, in three different ways: one cart moving into one at rest, two carts approaching each other at similar speeds, and a loaded cart running into an empty one. Each run was recorded by two motion sensors, so you have position against time for both carts through the whole event. You will pull four velocities out of each run — each cart before, each cart after — add up the two totals, and see how close they come. The interesting part is not that they agree. It is *how* they disagree, and whether the disagreement is big enough to mean anything.

Due see the Module 4 Lab folder on Brightspace | **Report** Full lab report — use the Lab Template | **Pairs with** M4 N8 | **Data** provided — this is a virtual lab

PERFORMANCE EXPECTATIONS what you should be able to do when this lab is written up

- **PE1** **State a sign convention for a one-dimensional system and apply it consistently** — including to a sensor that reports position along its own axis rather than yours, so that a velocity's sign means something physical and not merely which instrument recorded it.
- **PE2** **Extract a velocity from position-versus-time data** by fitting a straight line to a region where the cart is genuinely moving at constant speed, and state the fit window you used.
- **PE3** **Identify the interaction in the data without being told where it is** — recognising it as the interval over which both carts' slopes are changing, and excluding it from every fit.
- **PE4** **Compute the total momentum of a two-body system before and after an interaction**, $\Sigma p = m_A v_A + m_B v_B$, carrying the signs through, and quote the result with its units.
- **PE5** **Judge whether momentum was conserved against a stated criterion**, and recognise the case in which a percent difference is the wrong tool because its denominator is near zero — saying so rather than reporting the number it produces.
- **PE6** **Account for the direction of a discrepancy**, not merely its size — explaining what in the apparatus could remove momentum from the system and whether the data support that explanation.

Start from Note 8 — this lab is that note, measured

Note 8 is not background reading for this lab. It is the lab. The equation you wrote there — $p_{total,before} = p_A + p_B = m_A v_A + m_B v_B$, and the same sum with primes afterwards — is the entire analysis of this report. What the note could not give you is a set of real numbers that do not quite agree. That is what you have here, and the gap between "the law says these are equal" and "my two totals differ by this much" is the thing the report is about.

Where each part of Note 8 belongs in this report. The rubric awards points for these.

Report section	What Note 8 gives you
Introduction	The conservation law stated in your own words, and the equation $m_A v_A + m_B v_B = m_A v_A' + m_B v_B'$ with every symbol defined. State what "conserved" would look like in *this* measurement — a number you can compare, not a sentiment. End with a hypothesis you can later confirm or reject.
Experimental	The labelled setup drawing, redrawn by you, with **the positive direction marked on it**. State the sign convention and say explicitly how you handled the second sensor. Describe how the data were collected and why the carts never touch.
Analysis	Four velocities per trial from four straight-line fits, then the two totals and their difference. One worked sample calculation, by hand, with units, for each *type* of calculation you do.
Discussion	Note 8's argument for *why* the total cannot change — equal and opposite forces acting for the same time — checked against what you actually measured, including the trials where the agreement is worst.

Read or reprint the note here: Module 4 Note 8 — Momentum Conservation.

The physics you are using

One equation, applied three times. For two bodies on a line,

$$\Sigma p_{before} = m_A v_A + m_B v_B \qquad \Sigma p_{after} = m_A v_A' + m_B v_B'$$

and the law says those two are the same number. The symbol **Σp** is just shorthand for *the sum of every momentum in the system* — the same p_{total} you wrote in Note 8. What you actually report is the difference between them,

$$\Delta p = \Sigma p_{after} - \Sigma p_{before}$$

which the law says is zero and your data will say is merely small. Both halves of that sentence matter.

Sign convention here, and there is a trap in it. Momentum is a vector, and in one dimension a vector is a number with a sign. The two sensors do face each other from opposite ends of the track — so it is natural to assume that one cart's velocities must be flipped before they can be added to the other's. **In this data set they must not.** Cart B's positions are recorded as negative numbers precisely so that both columns already increase for motion in the same physical direction. A positive velocity means the same thing for both carts, and the two columns are ready to be added as they stand.

Do not take that on my word — it is checkable, and checking it is part of the lab. Look at Trial 1: cart A moves, stops, and then cart B starts moving. If both velocities carry the same sign, the momentum cart A lost is the momentum cart B gained, and the totals agree. If you flip cart B, cart A's forward momentum and cart B's would have to cancel — meaning the system was nearly at rest afterwards while a cart was visibly rolling down the track. One of those two readings describes what happened and the other does not. **Establish the convention**

this way, from the data, and say in your Experimental section how you established it. A sign error here does not make the totals slightly wrong; it makes them meaningless, and the tell is unmistakable — a trial that should nearly balance misses by more than the whole momentum you started with.

Predict first. In Trial 2 the two carts approach each other at nearly the same speed and bounce apart. Their masses are nearly equal too. What do you expect the *total* momentum of the system to be, before and after?

(A) Close to zero both times — and that is going to make the percent difference useless.

(B) Close to zero both times, so the percent difference will be excellent.

(C) Large before and large after, since both carts are moving fast.

(D) Zero before and nonzero after, because bouncing adds momentum.

Why that question is in this lab. Two nearly equal masses moving at nearly equal speeds in opposite directions have momenta that nearly cancel, so Σp is a small number both before and after. The law is satisfied just as strictly as in the other two trials — but a percent difference divides by that small number, and dividing a small absolute discrepancy by a small total produces an enormous percentage. **You will see something in the region of 60% on Trial 2, and it does not mean momentum was violated.** It means the tool is wrong for the job. Part III tells you what to report instead. Noticing this yourself, and saying so in the Discussion, is worth more than any of the arithmetic in this lab.

The apparatus

A level aluminium track with two low-friction carts on it. At each end sits an ultrasonic motion sensor, and the two face inward along the track. Each cart carries a small magnet, so the carts **never physically touch** — they repel each other through a short interaction at close range. This is deliberate: a magnetic interaction is *nearly* elastic and is gentle enough that nothing gets damaged, but every word of the momentum argument applies to it exactly as it would to a collision.

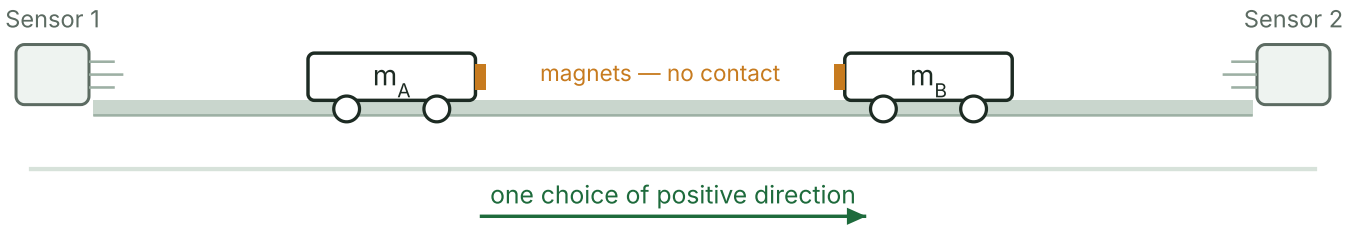


Figure 1. The track as the data were taken. Both sensors face inward, so they measure position along *opposite* axes. The positive direction drawn here is one choice — state yours, and stay with it.

The masses

Measured on a triple-beam balance and the same for every trial:

Symbol	What it is	Mass
m_A	Cart A, the one sensor 1 watches	0.5119 kg
m_B	Cart B, the one sensor 2 watches	0.5139 kg

Symbol	What it is	Mass
m_e	The extra bar, added to cart A in Trial 3 only	0.5319 kg

> **Trial 3 is the one where the masses are not what you expect.** The extra bar rides on cart A, so the moving mass in Trial 3 is $m_A + m_e = 1.0438$ kg, not 0.5119 kg. Using the bare cart mass there is the single most common way to get this lab wrong, and it will show up as a Trial 3 discrepancy of roughly 50%. Your Analysis must state which mass you used for each trial.

Part I — The data

Three runs, provided as Vernier Graphical Analysis files in the data bundle. Each contains three columns: time, cart A's position, and cart B's position, sampled every 0.05 s for 5 s.

The three trials. Each file holds one complete interaction.

File	What happens
Pt1 — Empty Carts, One at Rest	Cart A is pushed toward cart B, which sits still. After the interaction A is essentially stopped and B moves off. This is the cleanest of the three.
Pt2 — Empty Carts, Similar Speed	Both carts are launched toward each other at close to the same speed, and both rebound. This is the trial from the prediction above.
Pt3 — Cart A Heavy, Cart B at Rest	The extra bar is on cart A. A keeps moving forward after the interaction, more slowly, while B is pushed off faster than A is now going.

Open each file in Vernier Graphical Analysis. You should see two position traces. Before fitting anything, **look at the graph** and find the interval where both traces bend — that is the interaction, and it is the part you must fit *around*, never *through*.

Part II — Getting four velocities out of each run

On a position-versus-time graph a constant velocity is a straight line, and its slope *is* the velocity. Each cart has two of them per trial: one before the interaction and one after. Four fits per trial, twelve in all.

I. **Find the interaction.** Locate the time interval in which the slopes are changing. In these runs the magnets act over roughly a tenth of a second, so the bend is short but not instantaneous.

II. **Fit before and after, never across.** Select a region entirely on one side of the bend and apply a linear fit. Record the slope and the x-range you used — **the fit window is a required disclosure in your report**, on the same footing as the number it produced.

III. **Stay off the ends.** A cart that has reached the end of the track, or is still being pushed by hand, is not moving freely. Those regions are not part of the measurement.

IV. **Expect some velocities to be zero.** A cart at rest gives a flat line and a slope near zero. That is a real measurement, not a missing one, and it goes into the sum as 0.

> **Why the fit window is treated so seriously here.** It is the one choice in this lab that is yours alone, and two students analysing the same file can get different velocities purely by selecting different ranges. That is not a flaw to be hidden; it is the reason the window has to be reported. A result you cannot reproduce from the information in the report is not a result.

Part III — The two totals, and the gap between them

For each trial, with the signs carried through:

$$\Sigma p_{before} = m_A v_A + m_B v_B \qquad \Sigma p_{after} = m_A v_A' + m_B v_B' \qquad \Delta p = \Sigma p_{after} - \Sigma p_{before}$$

Report Δp in kg·m/s for all three trials. Then, **for Trials 1 and 3 only**, express it as a percentage:

$$\text{percent difference} = 100 \times \Delta p \,/\, \tfrac{1}{2}(|\Sigma p_{before}| + |\Sigma p_{after}|)$$

Trial 2, and what to report instead

In Trial 2 both totals are close to zero, so the denominator above is close to zero and the percentage it returns is large and meaningless. **Do not report a percent difference for Trial 2.** Report Δp in kg·m/s, and compare it against something that is *not* near zero — the momentum of a single cart, for instance, which is the scale of the quantities that went into the sum. State in the Discussion why you made that substitution. A report that quotes a 60% difference on Trial 2 without comment has misunderstood the measurement; a report that quotes it and explains why it is the wrong number has understood it exactly.

What counts as agreement

For Trials 1 and 3, a percent difference of **10% or less** is a confirmation. Larger than that and you have either a sign error, the wrong mass on Trial 3, or a fit window that has strayed into the interaction — check those three before concluding anything about physics. For Trial 2, judge Δp against the single-cart momentum scale and say plainly whether it is small.

Check your analysis

Enter the masses and your four fitted velocities for one trial to check the arithmetic. Your report still has to show the fits, the windows and a sample calculation worked by hand with units — this tool checks arithmetic, it does not replace the analysis.

● TOOL · TOTAL MOMENTUM BEFORE AND AFTER

Mass of cart A, including any extra bar (kg)

0.5119

Mass of cart B (kg)

0.5139

v_A before (m/s, signed)

e.g. 0.5491

v_B before (m/s, signed)

e.g. 0

v_A' after (m/s, signed)

e.g. 0.0103

v_B' after (m/s, signed)

e.g. 0.5269

Signs matter: a velocity entered without its sign will give a total that is wrong rather than merely imprecise. The kinetic energies are shown for the Additional Question on energy; they are not part of the required analysis.

Additional Questions

Choose three of the five below and answer those three fully. Name which three you chose. Answer them in a section of their own headed **Additional Questions**, in **sentences and paragraphs** with the working shown, never as a bare numbered list and never by restating the question before answering it. Depth on three beats coverage of five; this section is worth 8 points on the rubric.

AQ1 — the energy that went missing. Momentum is conserved in all three trials. Kinetic energy is not: compute ½mv² summed over both carts, before and after, for each trial, and you will find every one of them ends with less than it started. Report the three losses as percentages. Then answer the physics question: these carts never touch, so there is no crunch of deforming metal to blame — where did the energy actually go, and why does Note 8's argument for momentum conservation say nothing whatsoever about energy? Rank the three trials by fractional energy loss and say whether the ranking makes sense to you.

AQ2 — the sign convention, taken seriously. There are two different operations here and they are easy to confuse. *First:* redo Trial 2's analysis with the opposite choice of positive direction — every velocity flips, both totals flip, Δp flips, and the physical conclusion is unchanged. *Second:* flip cart B's velocities while leaving cart A's alone, which is what the opposed sensors tempt you into. Show that this one is **not** a change of convention at all but a corruption of the data, and compute how badly it fails — in Trial 2 it turns a Δp of about 0.02 kg·m/s into something larger than the total momentum the system ever had. Explain the general principle that separates the two cases, and state the test you would apply to a new data set to find out which situation you are in.

AQ3 — how much does the fit window matter? Pick one cart in one trial and fit its velocity three times, using three different windows: a generous one running right up to the interaction, a conservative one well clear of it, and one that deliberately includes a few points from inside the bend. Report the three slopes. Propagate each into Σp and say how much Δp moves. Then answer: is the disagreement between your two momentum totals larger or smaller than the uncertainty your own choice of window introduces? If it is smaller, what have you actually demonstrated about conservation of momentum — and what have you not?

AQ4 — the trial that breaks the percentage. Section III told you not to compute a percent difference for Trial 2. Justify that instruction mathematically rather than taking it on authority. Write the percent difference as a function of the denominator, show how it behaves as that denominator approaches zero while the numerator stays fixed, and compute what Trial 2's percentage would be. Then propose a better figure of merit for a system whose total momentum is near zero, defend it, and apply it to your own Trial 2 numbers.

AQ5 — predicting Trial 3 before measuring it. In Trial 3 cart A carries the extra bar and cart B starts at rest. Using only the masses and cart A's measured initial velocity, and assuming the interaction is perfectly elastic, derive expressions for both final velocities and compute them. Compare with what you actually measured. The agreement will be good but not perfect — quantify the gap and say what it tells you about how elastic these magnetic interactions really are. (The elastic two-body result is standard; derive it from conservation of momentum together with conservation of kinetic energy, and show your working.)

Prompts for your Discussion

These are **not** the Additional Questions. Work them into your **Discussion** section as part of your argument, in your own order. Do not answer them as a list, and do not limit yourself to them.

I. Momentum conservation is not approximately true in this experiment — it is a strict law, and what you measured are three numbers that are not quite zero. Which of your three Δp values is most likely to be a real physical effect rather than measurement scatter, and what in the data makes you say so?

II. The carts never touch. Does that make the interaction more elastic, less elastic, or does it not bear on elasticity at all? Your energy numbers, if you did AQ1, are evidence here.

III. Every Δp you measured has a sign. Are the three signs the same? If a systematic effect such as track friction or a slight slope were at work, what sign would you expect, and is that what you see?

IV. The track was levelled by eye. Estimate how much a slope of half a degree over the interaction distance would change your velocities, and say whether that is big enough to account for what you found.

V. Trial 1 is the cleanest of the three and Trial 2 the messiest by percentage. Is Trial 2 really the worst measurement, or only the worst-scaled one? Defend your answer with numbers.

VI. In a real two-car collision, would you expect momentum to be conserved to the same precision you measured here? What is different about that situation, and what would you have to include in the system to make the statement true?

Files for this lab

HANDOUT

This lab, as a PDF

The whole page, printable. The calculator only works on the web version.

REQUIRED

The three data files

Pt1, Pt2 and Pt3 for Vernier Graphical Analysis.

EXAMPLE

Example workbook

The raw data, a blank analysis sheet laid out for you, and one trial worked through.

GRADING

Lab 05 rubric

Read it before you write, not after.

SUPPORT

Report starter

Section guidance only now — the sentence stems have gone. See the note inside.

NOTE

M4 N8 — Momentum Conservation

The note this lab measures.

REFERENCE

Formatting data tables

Table captions, units in headers, significant figures.

References to get you started

Your references should include material on the conservation of momentum and on the software used to analyse the motion. Your textbook, this handout and the Module 4 note pages all count, and you need at least one source beyond the textbook. The note this lab measures is Module 4 Note 8, and the energy side of the story is in Module 4 Note 10.

1. [Author's names], *Journal or Book title*, edition, Volume number, page number, (year).

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